

# CFD-HW2

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## Outline

The objective of this assignment is to assess how liquid water may behave when forced through a multi-inlet & outlet elbow pipe with respect to CFD methods in Ansys Fluent. Furthermore, the simulation results will be compared to preliminary calculations (based off prescribed boundary conditions and fluid properties). Most steps carried out here were instructed per the provided Fluent tutorial.

The fluid dynamics will be simulated twice; one with the (provided) default geometry of the elbow, and one with diameter of one of the inlets increased slightly. This will (retrospectively) validate the results of the Fluent model.

Although too late at this point, because some items were missed in the previous assignment (due to my inability to follow given instructions), those items will be addressed here after the fact (for my own sake).

## Fluid Properties

- Density:  $\rho = 1000 \text{ kg/m}^3$
- Specific Heat:  $c_p = 4216 \text{ J/kg/K}$
- Thermal Conductivity:  $k = 0.677 \text{ W/m/K}$
- Viscosity:  $\mu = 8 \times 10^{-4} \text{ kg/m/s}$

## Boundary Conditions

- Implicit Geometric BCs from the Mesh/Geometry File
- Large Inlet
  - Velocity:  $\vec{v} = (0.4 \text{ m/s})\hat{x} + (0 \text{ m/s})\hat{y} + (0 \text{ m/s})\hat{z}$
  - Turbulence Intensity: 5%
  - Hydraulic Diameter: 100 mm
  - Temperature: 293.15 K
- Small Inlet
  - Velocity:  $\vec{v} = (0 \text{ m/s})\hat{x} + (1.2 \text{ m/s})\hat{y} + (0 \text{ m/s})\hat{z}$
  - Turbulence Intensity: 5%
  - Hydraulic Diameter: 25 mm
  - Temperature: 313.15 K
- Large Outlet
  - Turbulent Intensity (backflow): 5%
  - Hydraulic Diameter (backflow): 100 mm

## Preliminary Calculations

The mass flow rate through a cross section is given (in general) as  $\dot{m} = \rho Av$ , where  $\rho$  is the fluid density,  $A$  is the cross sectional area, and  $v$  is the fluid (normal) speed. Thus, the mass flow rates through the large and small inlets ( $\dot{m}_{iL}$  and  $\dot{m}_{iS}$ ) are computed as such:

$$\dot{m}_{iL} = (1000 \text{ kg/m}^3)(\frac{1}{4}\pi(0.1 \text{ m})^2)(0.4 \text{ m/s}) \approx 3.142 \text{ kg/s}$$

and

$$\dot{m}_{iS} = (1000 \text{ kg/m}^3)(\frac{1}{4}\pi(0.025 \text{ m})^2)(1.2 \text{ m/s}) \approx 0.589 \text{ kg/s}$$

The mass flow rate through the large outlet is  $\dot{m}_{oL} = \dot{m}_{iL} + \dot{m}_{iS} \approx 3.731 \text{ kg/s}$  by conservation of mass. Because we expect the fluid density to remain invariant throughout the elbow, the average fluid speed at the outlet can be computed as such:

$$v_{oL} = \frac{\dot{m}_{oL}}{\rho_{oL}A_{oL}} \approx \frac{3.731 \text{ kg/s}}{(1000 \text{ kg/m}^3)(\frac{1}{4}\pi(0.1 \text{ m})^2)} \approx 0.475 \text{ m/s}$$

Likewise, because we expect the thermal conductivity to remain invariant throughout the elbow, the average temperature at the outlet can be computed as such by conservation of energy:

$$T_{oL} = \frac{\dot{m}_{iL}T_{iL} + \dot{m}_{iS}T_{iS}}{\dot{m}_{oL}} \approx \frac{(3.142 \text{ kg/s})(293.15 \text{ K}) + (0.589 \text{ kg/s})(313.15 \text{ K})}{3.731 \text{ kg/s}} \approx 296.3 \text{ K}$$

## Procedure

Initially, as we were provided the elbow's geometry file, the first step was to import the file into the (Ansys) workbench. Then, a mesh had to be imposed onto the model in conjunction with referential (face) labels to run Fluent effectively using the meshing feature. Figure 1 depicts the mesh as follows:

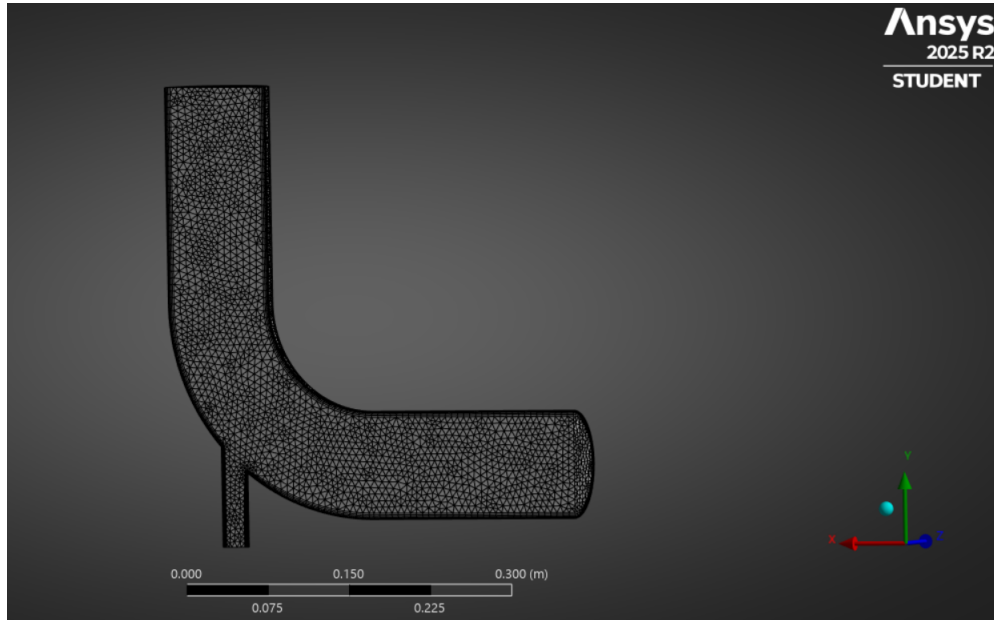


Figure 1: Meshing the Elbow in Ansys

Next, the model was set up in Fluent (accessed from the workbench). In Fluent, the  $k-\epsilon$  turbulence model was implemented. With the physics established, all that was left to do was provide the fluid properties and boundary conditions as prescribed. To be specific, liquid water (found in the material catalog) with its default properties was selected.

Once that was completed, a temperature monitor (with an associated plot) which monitors the large outlet was created. Likewise, a residual monitor for the fluid's velocity, energy, and other indicators was created (with an associated plot). Having post calculation residuals being small (e.g. energy residual  $< 10^{-6}$ ) indicates convergence of the model.

With all said and done, the simulation was ready to be run. As instructed, the number of iterations was set to 300 to validate convergence. The resultant plots are shown in figures 2 & 3.

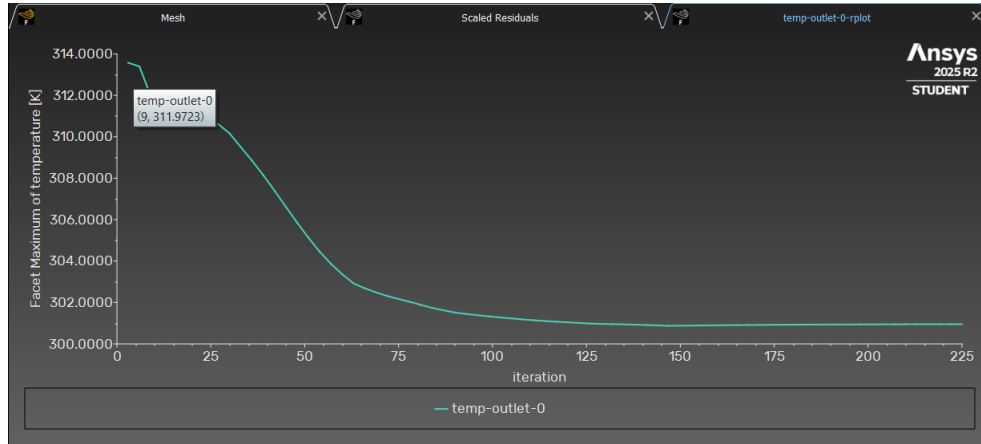


Figure 2: Iterated Maximum Temperature at the Outlet

Figure 2 demonstrates that after  $\sim 225$  iterations, the maximum temperature at the large outlet (boundary) settled to  $\sim 301$  K. This is (roughly) consistent with the predicted average temperature there (296.3 K).

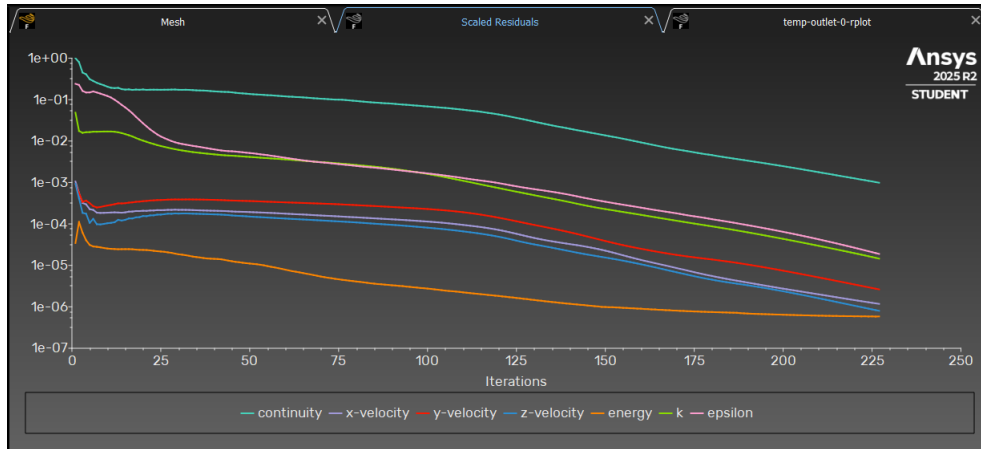


Figure 3: Residual Plots

Figure 3 demonstrates that the residuals did in fact go down a fair amount, with the energy residual in particular reaching sub  $10^{-6}$ . Overall, figures 2 and 3 indicate the model converged effectively.

With everything computed, the next item of interest was to display the results in CFD-post (workbench). Before entering CFD-post, two contours were created in Fluent which correspond with (magnitude of) velocity and temperature on the symmetry plane of the elbow; effectively color (scalar) fields. Then, in CFD-post, the contours were implemented for display purposes, which are shown in figures 4 & 5.

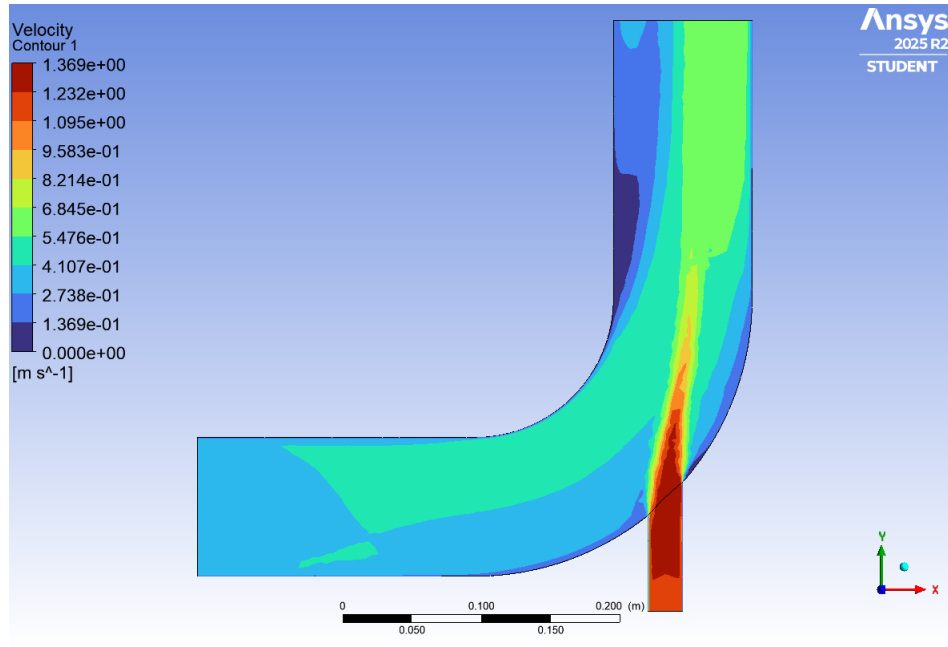


Figure 4: Fluid Speed (color) Field on Symmetry Plane

Upon inspection, the average fluid speed exiting the large outlet is between 0.137 m/s (min) and 0.548 m/s (max). The average of these two values comes out to (roughly) 0.343 m/s. With respect to orders of magnitude, this is inline with the predicted value of 0.475 m/s, which validates the models' accuracy.

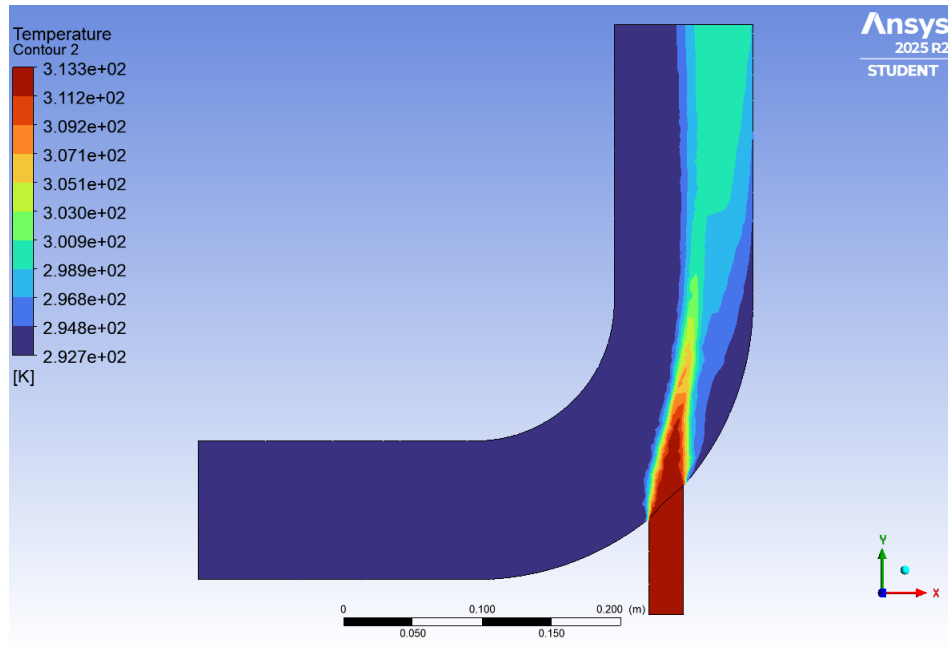


Figure 5: Fluid Temperature (color) Field on Symmetry Plane

At this point, per the tutorial, this process was replicated with one modification, but in a streamlined manner. To be specific, the Fluent system was duplicated, and the diameter of the smaller inlet was changed to 38 millimeters in the (new) geometry file via the Ansys design modeler. The (new) mesh was also updated.

After going back into fluent (changing the hydraulic diameter of the small inlet to 38 mm), the model was reinitialized, and rerun. To avoid redundancy, here it will be reported (without plots) that the residuals in fact decayed to equivalent orders of magnitude as to before (in 300 iterations), and the max temperature at

the outlet settled to (roughly) 306 K after (at least) 225 iterations.

With the hard part out of the way, it was time to compile and display the final results from each analysis. The two solutions from both analyses were combined into a single location in the workbench (illustrated in figure 6), where the respective contour fields were displayed in CFD-post. The results from both simulations on the symmetry planes are (likewise) shown below (from CFD-post) in figures 7 & 8.

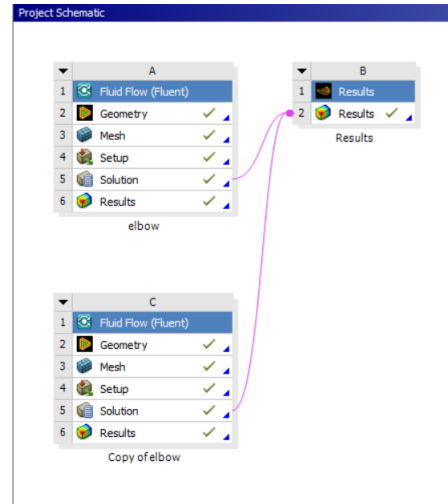


Figure 6: Complete Workbench

Note that no errors were encountered in this attempt relative to my attempt at the previous assignment, and (thankfully) hence the uniform representation of check marks displayed in the workbench. The errors being alluded from before likely originated me fooling around too much.

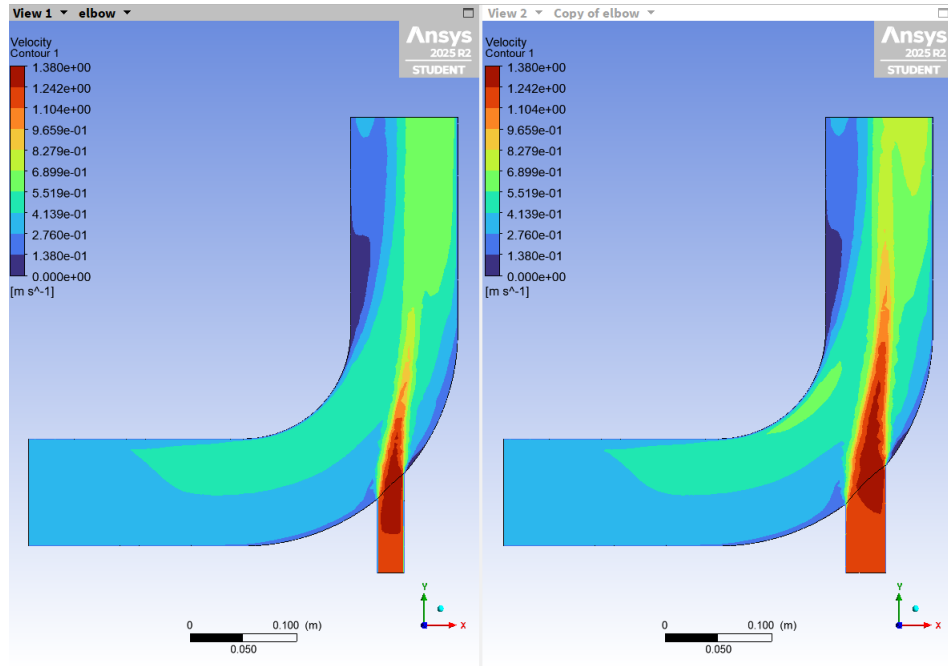


Figure 7: Fluid Speed Comparison (old on left and new on right)

It is apparent in figure 7 that the average fluid speed inside the elbow increases overall when the (smaller) inlet is enlarged. This makes sense being that the fluid velocity at the boundary was not altered, but the inlet area (i.e. the amount of influenced fluid) was. Another item to highlight in both cases is that large speed gradients remain intact along the  $x$ -axis, whilst on the other hand, the speed gradient gradually disperses

along the  $y$ -axis. This phenomena would be quite cumbersome to observe by hand (i.e. without computer assistance).

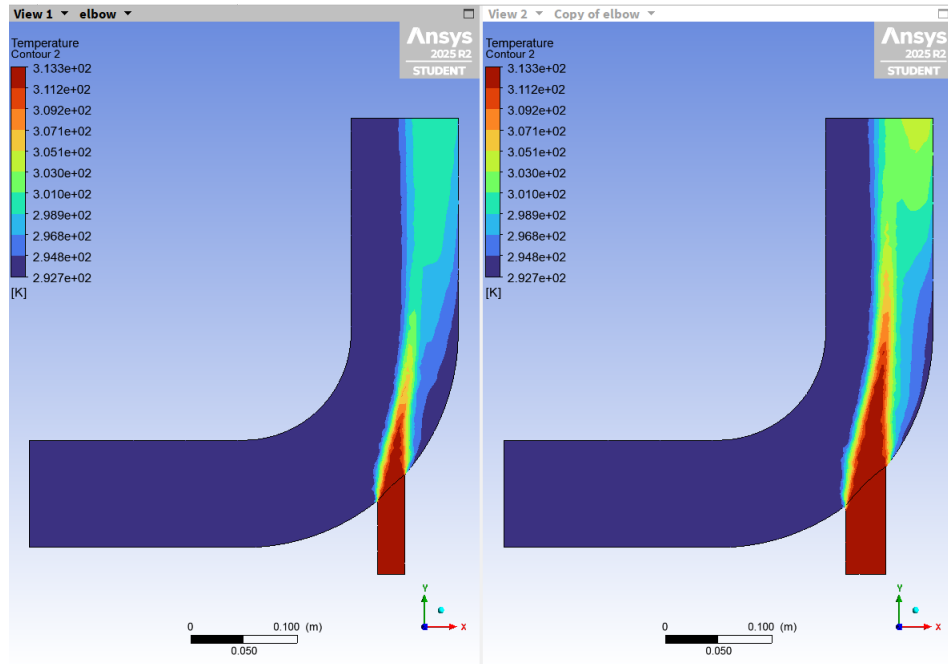


Figure 8: Fluid Temperature Comparison (old on left and new on right)

The analysis from figure 7 follows over here almost entirely . The most distinct exception relates to how the lateral (temperature) gradient behaves. That is, the temperature is (roughly) uniform before (along the  $x$ -axis) the fluid interacts with the small inlet stream, which reflects the fluid's thermal conductivity.

## Continued Procedure (HW1)

Now to tend to what was missed in the previous assignment. To start, in the same workbench established here, another duplicate of the original Fluent system was created for organizational purposes. In this duplicate, the simulation was run again. The (mass) flux results through the inlets and outlet were computed, and are shown in figure 9 (from the in-Fluent console).

| Console        |               |
|----------------|---------------|
| Mass Flow Rate | [kg/s]        |
| large-inlet_   | 1.5641563     |
| large-outlet   | -1.8533344    |
| small-inlet    | 0.28918197    |
| Net            | 3.8930415e-06 |

Figure 9: Flux Results

Next up, a colored (with respect to speed) fluid velocity field on the symmetry plane will be created. The results were (later) tended to in CDF-post, and are shown below in figure 10. Although a velocity field was generated in the previous assignment, it was not treated properly in CDF-post. That will be rectified here (again, for my own sake).

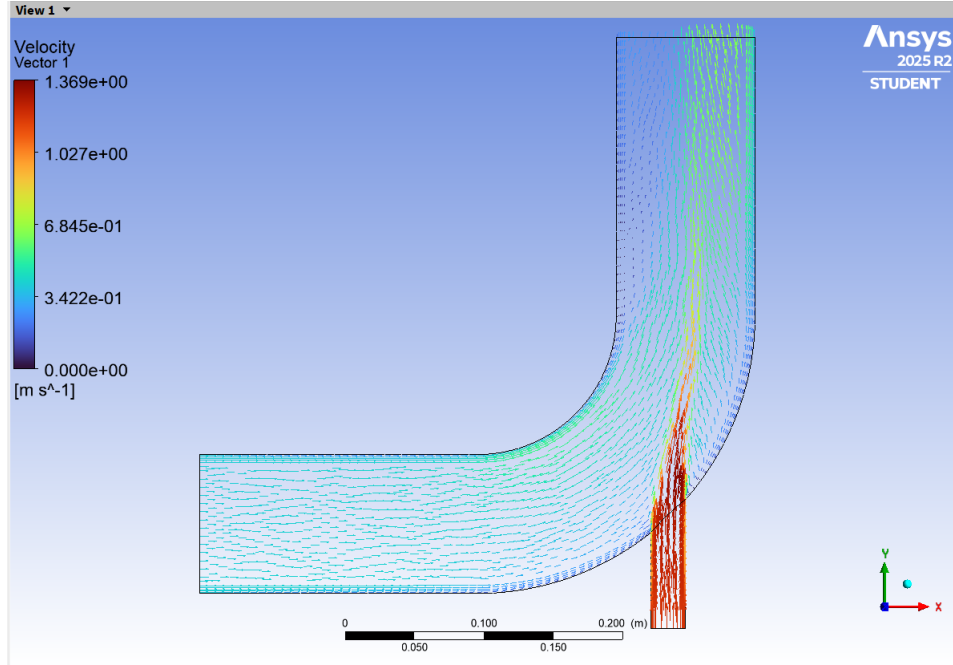


Figure 10: Velocity Field

In order to generate a temperature profile along the large outlet, an iso-surface had to first be generated at that boundary location. Then a plot linked to the temperature measured along the surface was created. The respective plot is displayed in figure 11.

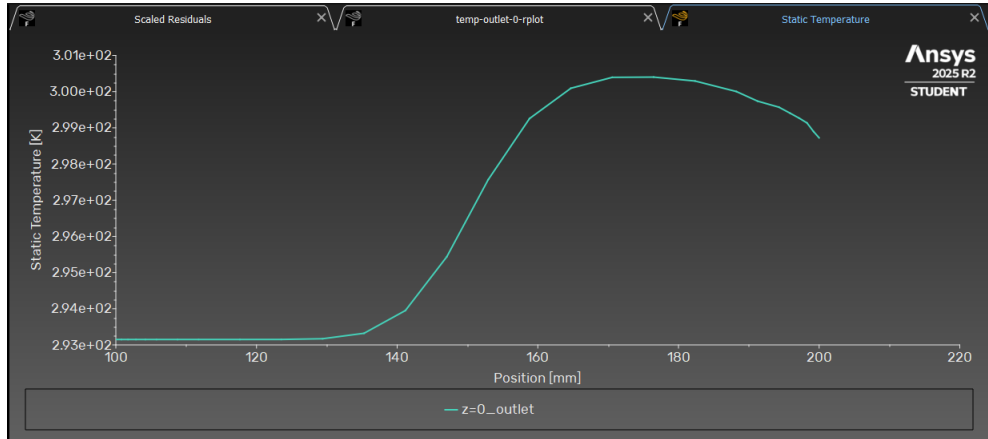


Figure 11: Outlet Temperature Profile

At a glance, Figure 11 matches the results from figures 5 & 8 (at the large outlet boundary). If allotted more time, the data would be extracted, and numerically integrated to estimate the average temperature along the profile to further validate the model.

The subsequent agenda item was to construct a custom scalar field on the symmetry plane. It was instructed to generate a field of the form  $f = \frac{1}{2}\rho||\vec{v}||^2$ . The resultant (color field) plot is provided here in figure 12. Note that despite my efforts, I couldn't get the contour pertaining to  $f$  to be displayed inside CFD-post. So the default display will have to do for now.

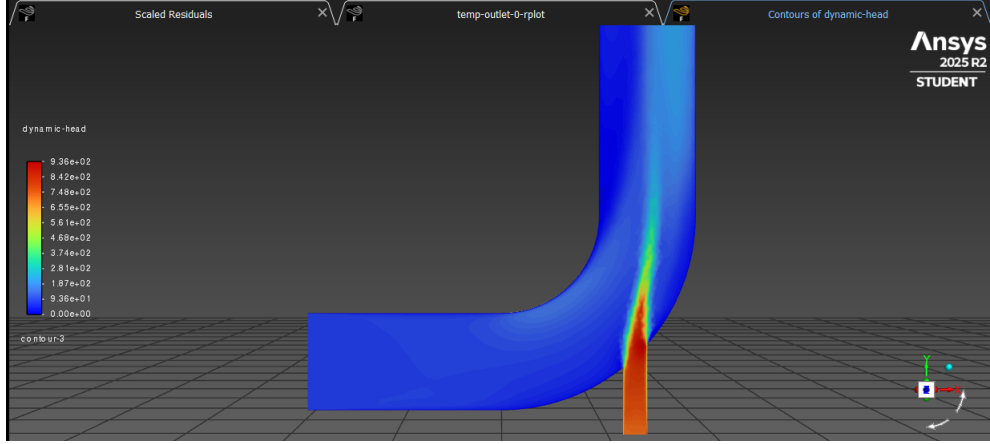


Figure 12: Custom Field

Although not explicitly mentioned, the simple scheme for the numerical CFD method was used throughout. Now, the coupled method (with 90 iterations instead of 300) will be run. The system was reinitialized, and rerun. The residual plot is given below in figure 13.

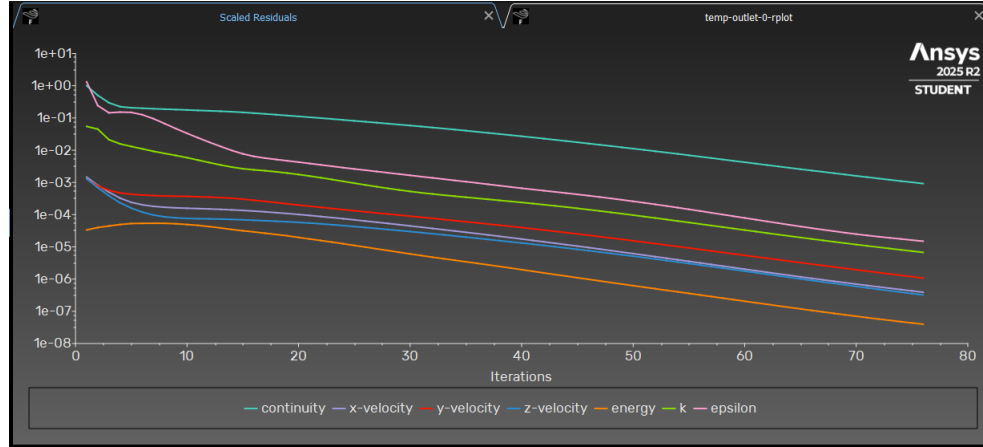


Figure 13: Coupled Residual Plot

By inspection, once again, the residuals (in particular, energy) meet the typical convergence specs ( $< 10^{-6}$ ). And again, to reduce redundancy, without showing the plot, it will be reported that the max temperature along the large outlet boundary remarkably settled down to  $\sim 301$  K after (at least) 75 iterations.

The final concept to explore here will be adapted meshes. The purpose of an adapted mesh is to provide higher (meshing) concentration where the large gradients (in some variable) were already calculated. This increases the model's resolution where it needs it the most. This procedure was implemented based off the temperature gradient in the symmetry plane as depicted in figure 14.

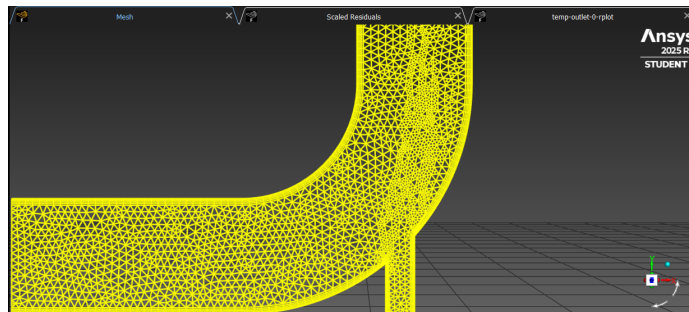


Figure 14: Adapted Mesh



In figure 14, although it's a little hard to see, the mesh is more concentrated where the hotter fluid is being dispersed (from the small inlet) just as desired.

With the new mesh implemented, the simulation was ready to be rerun, and so it was. The residuals are shown in figure 15. The max temperature at the large outlet settled down to 301 K for reference.

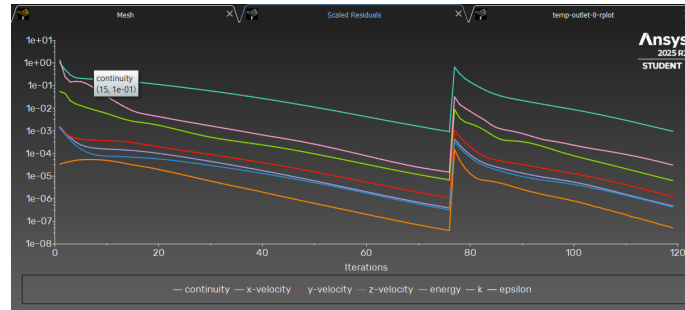


Figure 15: Adapted Residuals

The results from before and (now) after the mesh adaptation can be analyzed. Figure 16 illustrates the temperature profiles at the outlet boundary in each respective case.

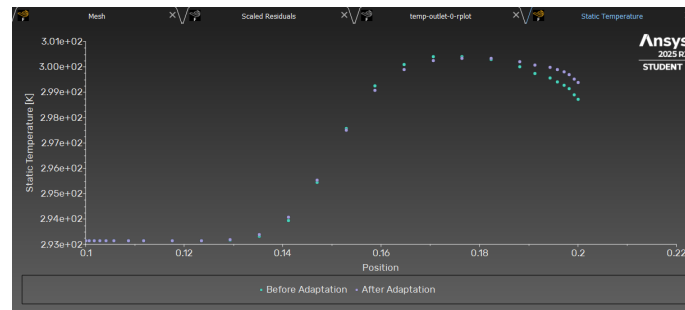


Figure 16: Adapted Versus Unadapted Temperature Profiles

The temperature profiles are very similar as shown in figure 16. The biggest difference occurs around the right side ( $x$ -axis) of the outlet. There, the unadapted profile dips down slightly faster than the adapted one. However, overall, the results are essentially the same, which is expected.

## Conclusionary Remarks

Retrospectively, this assignment (in conjunction with the previous one) provided a very helpful introductory overview of Ansys Fluent. It involved boundary condition implementation, residual (convergence) analysis of CFD models, field variable visualizations, post (display) analysis, alternative simulation methodology approaches, and so on.

Being that the constructed model behaved internally consistent, and consistent with preliminary (physics) calculations, it's safe to say at this point that the CFD model(s) presented here indeed accurately depict fluid dynamics in an elbow pipe to (at least) some extent. This motivates further investigation into these modeling techniques for futuristic ambitions regarding physics computations.